MATH I (Fall 2026)

WEEK 6: INDEPENDENT STUDY ASSIGNMENTS (Homework)

About the Independent Study Assignments

The Independent Study Assignments are homework problems and assignments that will help you learn the week's content more deeply by solving problems and expanding your knowledge and skills. *Your answers will not be submitted and will not be graded*. You may solve the problems individually or collaborate with your fellow students or ask the assistance from an AI tool (to help you clarify the concepts, not providing the solution without you thinking about it first).

Assignment 1. [*Optional assignment*]

The following is an *optional assignment – you can engage with it if you are familiar with AI-assistants, or if you want to start learning what AI-assistants can do for you in a context-specific domain*. The exercise is as follows:

Interact with an AI-assistant (such as ChatGPT, Claude, DeepSeek, Gemini, etc.), by structured conversation, through the following steps:

Think of the most important points of the lecture's content and ask the AI-assistant to evaluate your answer (you may ask the AI-assistant to evaluate it (usually, they do the evaluation in terms of content and clarity, but you can also ask for depth and complexity, or other criteria you may choose). The AI-assistant will not grade your input but it will provide content and perspectives to help you better understand the material.

If you find wrong outcomes, discrepancies or contradictory statements, converse with the AI-assistant to mention these points, asking for explanations, to resolve them. If you cannot resolve the wrong outcomes, please consult the instructor.

Assignment 2:

Study well **all** the problems, exercises, and examples that were presented in Lecture and Practical Session.

Assignment 3:

Study Examples 3.34 and 3.35 in the textbook (OpenStax Calculus 1).

Assignment 4:

Solve the following problems (they are from the textbook, in pages 196, 210, 211, 212, 232,

- [x] 3.4 A rock is dropped from a height of 64 feet. Its height above ground at time t seconds later is given by $s(t) = -16t^2 + 64, 0 \le t \le 2$. Find its instantaneous velocity 1 second after it is dropped, using Equation 3.5.

- [x] 3.8 Find values of a and b that make $f(x) = \begin{cases} ax + b \text{ if } x < 3 \\ x^2 \text{ if } x \ge 3 \end{cases}$ both continuous and differentiable at 3.

- [x] 3.9 Find $f''(x)$ for $f(x) = x^2$.

- [x] 3.10 For $s(t) = t^3$, find $a(t)$.

- [x] 3.21 Given $f(10) = -5$ and $f'(10) = 6$, estimate $f(10.1)$.

Assignment 5

Consider

$$f(x) = \begin{cases} 2x + 1, & x < 0, \\ x^2 + 1, & x \ge 0. \end{cases}$$

a. Check continuity at 0.
b. Find $f'_-(0)$ and $f'_+(0)$.
c. Determine whether f is differentiable at 0.
d. Describe the graph near the joining point.

Assignment 6

Determine whether each statement is true or false. Give a reason.

a. If f is differentiable at a, then f is continuous at a.
b. If f is continuous at a, then f is differentiable at a.
c. If $f'(a) = 0$, then f must be constant.
d. If both one-sided derivatives exist and are finite, then $f'(a)$ exists.
e. A function can have a vertical tangent at a point where it is continuous.

[Optional Assignment (for those who like racecars and racetracks)] Study the student project in page 226 of the textbook (OpenStax Calculus I). Understand its objective and the questions asked. Try to answer the questions.

[The student project is presented below]

STUDENT PROJECT

Formula One Grandstands

Formula One car races can be very exciting to watch and attract a lot of spectators. Formula One track designers have to ensure sufficient grandstand space is available around the track to accommodate these viewers. However, car racing can be dangerous, and safety considerations are paramount. The grandstands must be placed where spectators will not be in danger should a driver lose control of a car (Figure 3.20).



Figure 3.20 The grandstand next to a straightaway of the Circuit de Barcelona-Catalunya race track, located where the spectators are not in danger.

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Safety is especially a concern on turns. If a driver does not slow down enough before entering the turn, the car may slide off the racetrack. Normally, this just results in a wider turn, which slows the driver down. But if the driver loses control completely, the car may fly off the track entirely, on a path tangent to the curve of the racetrack.

Suppose you are designing a new Formula One track. One section of the track can be modeled by the function $f(x) = x^3 + 3x^2 + x$ (Figure 3.21). The current plan calls for grandstands to be built along the first straightaway and around a portion of the first curve. The plans call for the front corner of the grandstand to be located at the point $(-1.9, 2.8)$. We want to determine whether this location puts the spectators in danger if a driver loses control of the car.

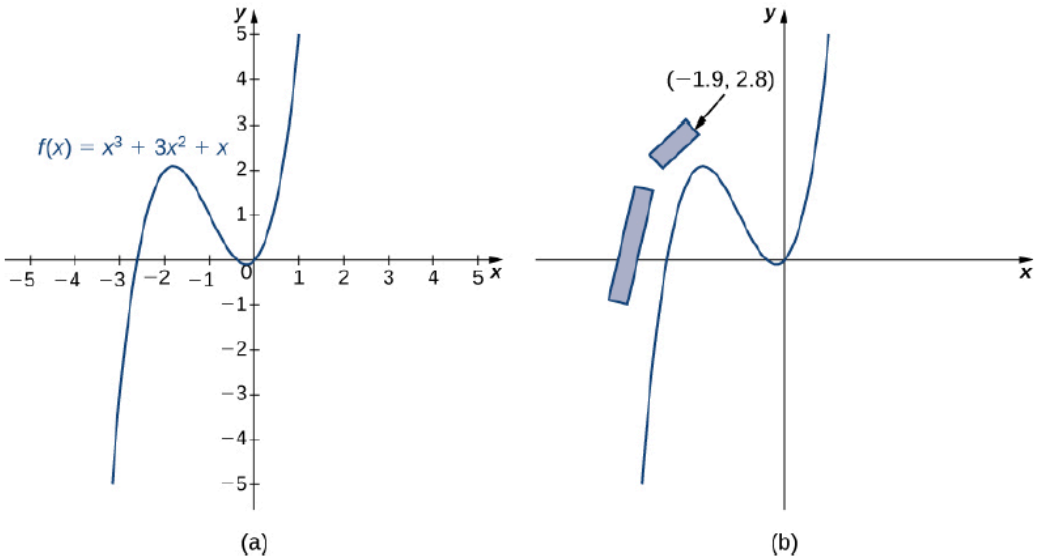


Figure 3.21 (a) One section of the racetrack can be modeled by the function $f(x) = x^3 + 3x^2 + x$. (b) The front corner of the grandstand is located at $(-1.9, 2.8)$.

1. Physicists have determined that drivers are most likely to lose control of their cars as they are coming into a turn, at the point where the slope of the tangent line is 1. Find the (x, y) coordinates of this point near the turn.
2. Find an equation of the tangent line to the curve at this point.

3. To determine whether the spectators are in danger in this scenario, find the x-coordinate of the point where the tangent line crosses the line $y = 2.8$. Is this point safely to the right of the grandstand? Or are the spectators in danger?
4. What if a driver loses control earlier than the physicists project? Suppose a driver loses control at the point $(-2.5, 0.625)$. What is the slope of the tangent line at this point?
5. If a driver loses control as described in part 4, are the spectators safe?
6. Should you proceed with the current design for the grandstand, or should the grandstands be moved?

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